Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY

9700/19

Paper 4 A Level Structured Questions – Principles of Genetic

MARK SCHEME

Maximum Mark: 113

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of 10 printed pages.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:
• the specific content of the mark scheme or the generic level descriptors for the question
• the specific skills defined in the mark scheme or in the generic level descriptors for the question
• the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always whole marks (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded positively:
• marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
• marks are awarded when candidates clearly demonstrate what they know and can do
• marks are not deducted for errors or omissions
• answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question.

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles & Abbreviations

1 Context and scientific use of keywords: marks should not be awarded if keywords are used incorrectly.
2 Contradictory statements: credit should not be awarded for any correct statement that is contradicted within the same question part.
3 Calculation guidance: correct answers to calculations should be given full credit even if there is no working, unless stated otherwise.
Abbreviations: ; separates marking points | **/** alternative answers | **R** reject | **A** accept | **AW** alternative wording | **underline** essential word | **max** maximum mark | **ecf** error carried forward | **I** ignore | **AVP** alternative valid point.

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Question	Answer	Marks
1(a)	• DNA that contains DNA / genetic material from two or more different sources / organisms ; ***A*** *artificial DNA formed by combining sequences from different species*	1
1(b)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	3
1(c)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	2
1(d)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	1
2(a)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	1
2(b)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	3

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Question	Answer	Marks
2(c)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
2(d)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	1
3(a)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	1
3(b)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
3(c)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
3(d)	• valid scientific hypothesis / plausible biological explanation ; • application of syllabus knowledge to novel scenario ; ***A*** *reasonable alternative suggestions supported by biological principles*	2

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Question	Answer	Marks
1(a)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**2**
1(b)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**4**
1(c)	• valid scientific hypothesis / plausible biological explanation ; • application of syllabus knowledge to novel scenario ; ***A*** *reasonable alternative suggestions supported by biological principles*	**2**
1(d)	• detailed account of sequence of events / structural features observed ; • specific factual details / trends / data points cited ; • comprehensive description addressing all aspects of the process / observation ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *descriptive points in any logical order*	**5**
2(a)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**2**

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Question	Answer	Marks
2(b)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**2**
2(c)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**2**
2(d)	• accurate biological statement meeting the syllabus learning objective ; • supporting detail / elaboration of biological concept ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *acceptable syllabus alternative [AW]*	**6**
3(a)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**2**
3(b)	• correct biological term / structure identified ; • further elaboration of biological principle / detail of mechanism ; ***R*** *ambiguous terminology*	**2**

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Question	Answer	Marks
3(c)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**3**
3(d)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct substitution of values from diagram / text into formula ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	**3**
4(a)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**2**
4(b)	• valid scientific hypothesis / plausible biological explanation ; • application of syllabus knowledge to novel scenario ; ***A*** *reasonable alternative suggestions supported by biological principles*	**1**
4(c)	• valid scientific hypothesis / plausible biological explanation ; • application of syllabus knowledge to novel scenario ; ***A*** *reasonable alternative suggestions supported by biological principles*	**1**
4(d)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**2**

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Question	Answer	Marks
1(a)	• correct biological term / structure identified ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***R*** *ambiguous terminology*	4
1(b)	• valid scientific hypothesis / plausible biological explanation ; • application of syllabus knowledge to novel scenario ; ***A*** *reasonable alternative suggestions supported by biological principles*	1
1(c)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	1
1(d)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
2(a)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	8

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Question	Answer	Marks
2(b)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; • further elaboration of biological principle / detail of mechanism ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**6**
3(a)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; • further elaboration of biological principle / detail of mechanism ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	**4**
3(b)	• primary structure: sequence of amino acids joined by peptide bonds ; • secondary structure: folding into alpha-helix or beta-pleated sheet held by hydrogen bonds ; • tertiary structure: further 3D compact folding held by hydrogen, ionic, disulfide bonds and hydrophobic interactions between R-groups ; • quaternary structure: association of two or more polypeptide chains (and prosthetic groups) ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *fibrous vs globular structural distinctions*	**11**
4(a)	• accurate biological statement meeting the syllabus learning objective ; • supporting detail / elaboration of biological concept ; • further elaboration of biological principle / detail of mechanism ; ***A*** *acceptable syllabus alternative [AW]*	**3**
4(b)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	**2**

Question	Answer	Marks
4(c)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	**6**
4(d)	• PCR: thermal cycling to amplify target DNA: denaturation (95°C), primer annealing (55°C), extension (72°C) via Taq polymerase ; • restriction enzymes / endonucleases cut DNA at specific palindromic recognition sites producing sticky ends ; • DNA ligase joins sugar-phosphate backbone forming recombinant plasmid / DNA ; • plasmids act as vectors to transfer foreign gene into host cells (bacteria) ; • marker genes (e.g. GFP / antibiotic resistance) used to identify successfully transformed cells ; ***A*** *gel electrophoresis separates fragments by size towards positive anode*	**3**

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